

1. The question

Market structure (reform calendar in one paragraph).

Who actually sets the price (NOT a priori CCGT).

Why this matters.

Outcomes we care about, and why.

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2. Variation across hours: where granularity has economic content

Intuition.

Classifier.

What we see.

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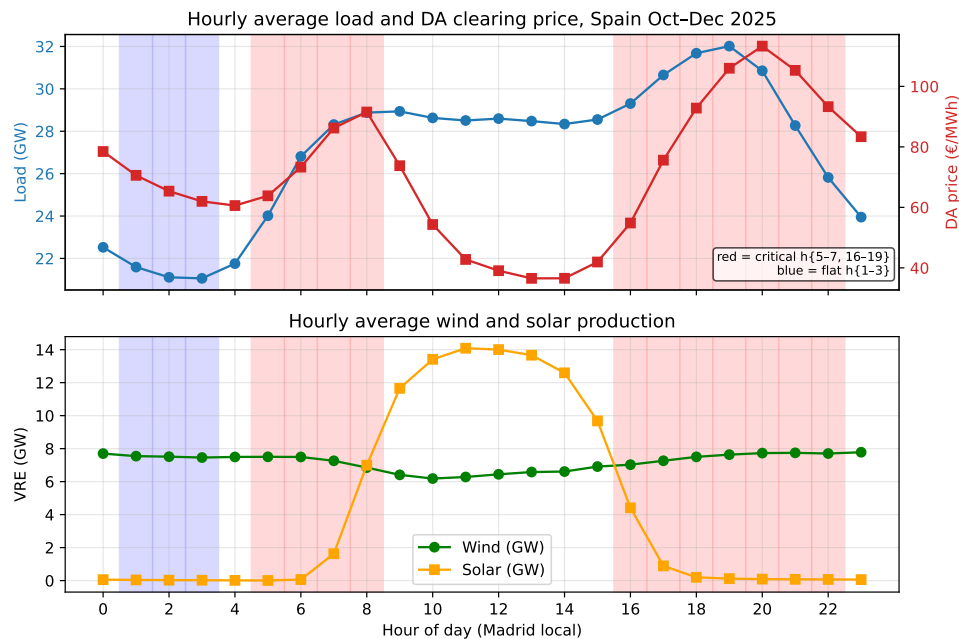


Figure 1:

Validation.

3. Variation across firms and technologies: how do CCGTs bid?

The object.

Metric: D_w .

Decomposition: Δp vs Δq .

What we see (post-MTU15-DA, critical hours, CCGT, Oct–Dec 2025).

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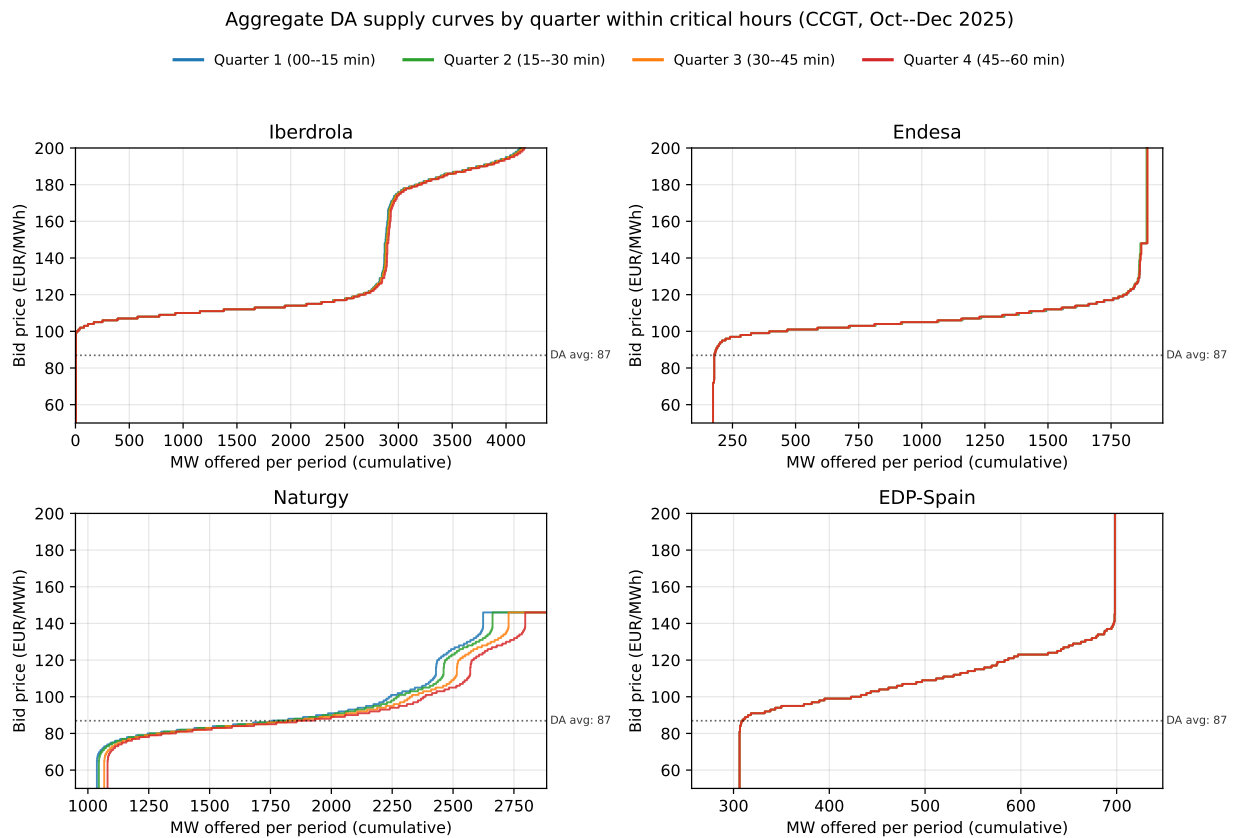


Figure 2:

Operational vs strategic decomposition.

4. The 2025 regulatory shock: blackout and reinforced operation

The blackout (28 April 2025).

What “reforzada” does in practice.

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What this lets us claim.

5. Reform windows we use, and outcomes we measure in each Strategy in one sentence.

The reform calendar and our windows.

Reform	Date	Window and why

Outcomes we measure in each window.

Outcome	What it tells us / why it matters

6. MTU15 effect: critical/flat identification (ID15 and DA15)

Strategy. Tight pre/post windows around each reform date, so reforzada level is constant within each arm; the (critical – flat) differential across the cutover strips common date-level shocks. Critical $h \in \{5, \dots, 8, 16, \dots, 22\}$ have non-trivial within-hour residual demand variance (522–1303 MW); flat $\{1, 2, 3\}$ is the within-day control (336–357 MW); other hours dropped.

Reform	Pre window	Post window	Reforzada
MTU15-IDA	2024-12-19 → 2025-03-18 (3 mo)	2025-03-19 → 2025-04-27 (40 d, clean)	absent both
MTU15-DA	2025-07-01 → 2025-09-30 (3 mo)	2025-10-01 → 2025-12-31 (3 mo)	active both, constant

6.1 Specifications

u unit; d date; p period (hour pre-, quarter post-MTU15); h clock-hour; s IDA session; T_r reform date. SEs clustered by date throughout; sample restricted to critical and flat hours.

A. Per-curve ($Y \in \{\sigma_p, N_{\text{eff}}\}$, in-band $|p_{\text{bid}} - \text{MCP}| \leq 140$):

$$Y_{u,d,p[s]} = \alpha + \alpha_u + \beta \mathbf{1}\{d \geq T_r\} + \xi \mathbf{1}\{h \in \mathcal{C}\} + \theta \mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon$$

B. Aggregate ($P \in \{\text{OMIE clearing price, DA cleared MW}\}$):

$$P_{d,p[s]} = \delta_d + \xi \mathbf{1}\{h \in \mathcal{C}\} + \theta \mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon \quad (\text{date FE absorbs post})$$

C. Within-hour dispersion, post-only. For $D \in \{D_{\text{price}} = \text{sd}_q \bar{p}_q, D_{\text{qty}} = \text{cv}_q m_q\}$ across the 4 within-hour quarters:

$$D^{(\cdot, d, h)} = \delta_d + \theta \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon$$

6.2 Results

Spec	Outcome (DiD θ)	Subsample	MTU15-IDA	MTU15-DA
A	σ_p (EUR/MWh)	CCGT	2.45*** (0.58)	1.30* (0.75)
A		Hydro	1.17*** (0.44)	1.69*** (0.29)
A		Hydro pump	0.25 (0.28)	-0.63* (0.36)
A	N_{eff}	CCGT	0.79*** (0.30)	0.44*** (0.06)
A		Hydro	0.21*** (0.06)	-0.04 (0.04)
A		Hydro pump	0.18** (0.08)	-0.05 (0.03)
B	Clearing price (EUR/MWh)	system	-11.18*** (3.75)	+26.70*** (2.54)
B	DA cleared MW	system	—	-2,115*** (259)
C	D_{price} (EUR/MWh)	per unit	0.39*** (0.05)	1.35*** (0.11)
C	D_{qty} (CV)	per unit	0.024*** (0.003)	0.033*** (0.002)
C	SD_{price} (EUR/MWh)	system	4.55*** (0.40)	3.97*** (0.37)

Notes: Standard errors in parentheses, clustered by date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Spec A: per-tech subsample regressions, unit FE absorbed via within-unit demeaning. Spec B: date FE absorbs post; DiD without FE: ID15 -11.32, DA15 +26.70 (unchanged). Spec C: post-only with date FE via within-date demeaning. Source: `results/regressions/bid/mtu15_critical_flat/`.

Reading (gaps for prose).

- σ_p : positive significant DiD in both reforms for CCGT and Hydro — MTU15 widens the in-band price ladder in critical hours. Per-tech heterogeneity: CCGT-led for ID15; Hydro-led for DA15 on σ_p , CCGT-led on N_{eff} .
- Prices: OPPOSITE signs across reforms. ID15 critical-hour IDA prices fall EUR 11/MWh more than flat (granular intraday absorbs scarcity peaks); DA15 critical-hour DA prices rise EUR 27/MWh more (granular day-ahead pricing extracts higher rents in scarce quarters).
- Quantity: DA15 critical-hour cleared MW falls -2,115 MW more than flat. Joint with the +EUR 27 price effect = textbook withholding signature.
- Spec C dispersion (post-only): every outcome is 3–7 $\times$ larger in critical than flat hours — direct confirmation that MTU15 unlocks discrimination only where within-hour variance exists.

6.3 Robustness (one-line each)

(All in `results/regressions/bid/mtu15_critical_flat/`.) (i) **Date FE** (Spec B): DiD unchanged $\Rightarrow$ not driven by common date-level shocks. (ii) **Midday** {11, ..., 14} **falsification** (Spec A pooled): σ_p DiD +1.10*** ID15 and +2.26*** DA15 — not zero but $\sim 65\%$ of the critical magnitude; consistent with midday's intermediate within-hour variance, reads as a dose-response in variance rather than a placebo failure. (iii) **Per-CCGT-firm** (IB/GE/GN/HC): large heterogeneity (e.g. ID15 IB σ_p +6.46***, GN N_{eff} +1.84***); DA15 GN headline σ_p +35.0 is a level-shift artefact (flat-hour mean collapsed 40.7 $\rightarrow$ 7.6) and should be flagged, not read as within-hour MTU15. **To do:** extended pre windows; same-calendar-month; CCGT scarcity-tier split.